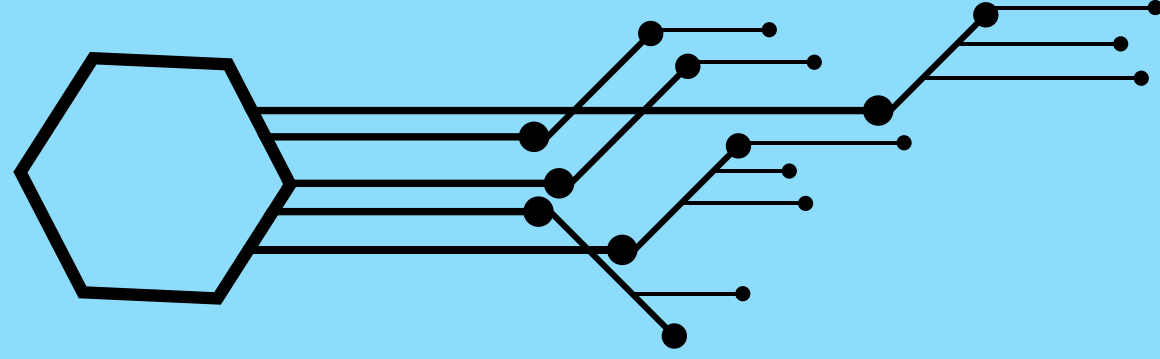


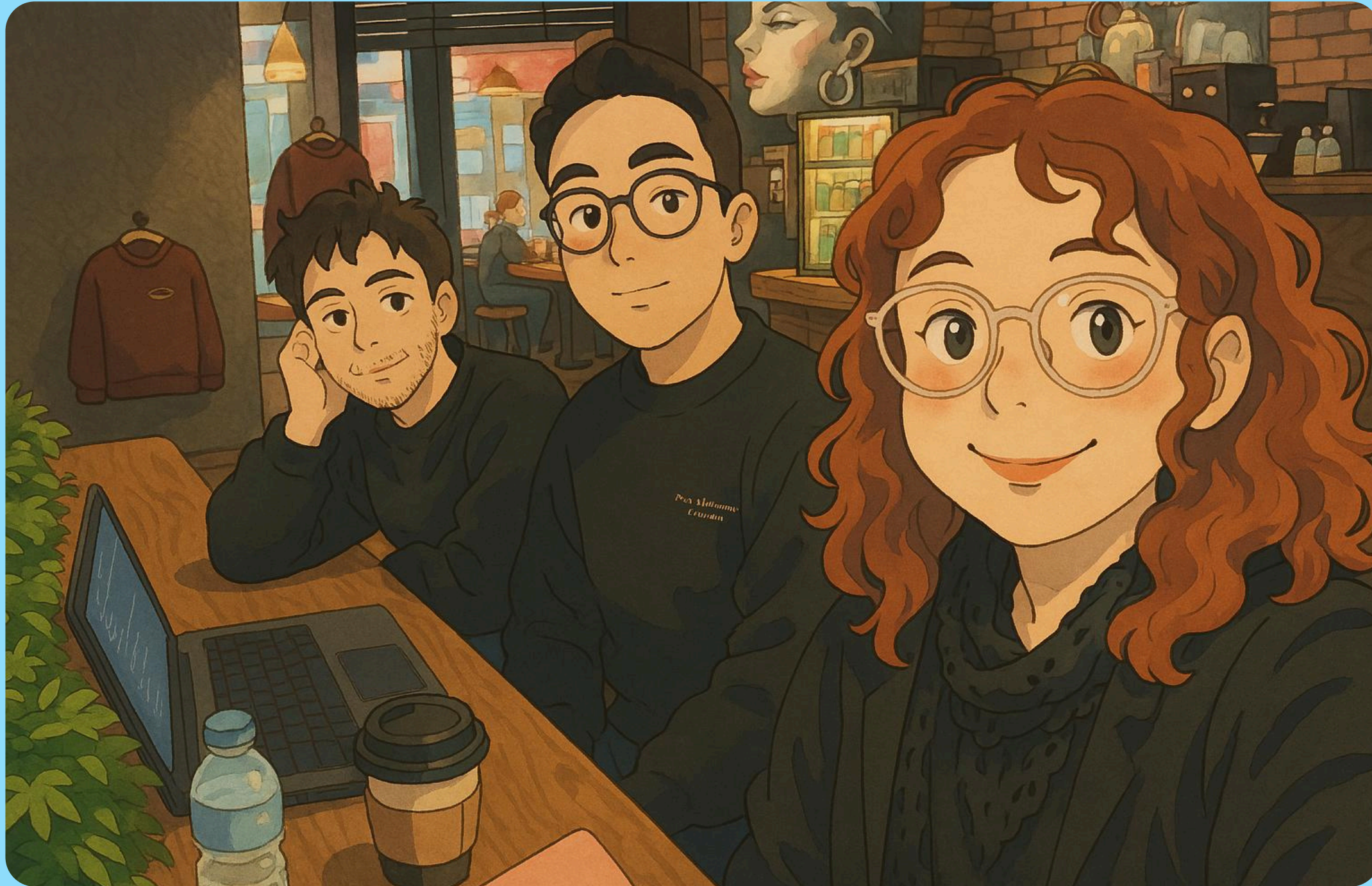
AI-Human Face Distinction

***CENG3522 APPLIED MACHINE
LEARNING PROJECT
2024-2025 SPRING SEMESTER***

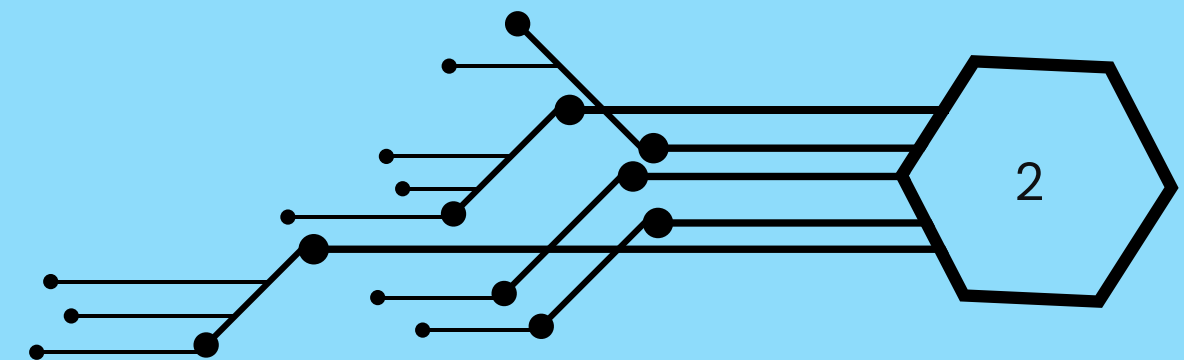




OUR TEAM



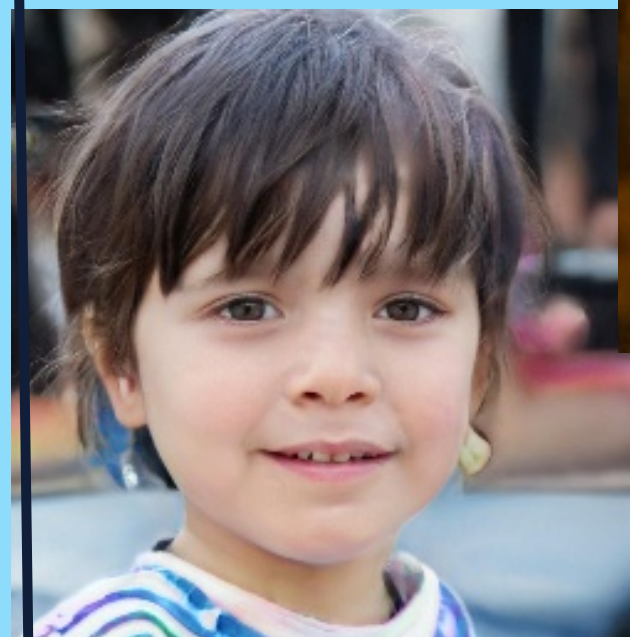
- *Merve Kaldırım*
mervekaldirim@posta.mu.edu.tr
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halilefecakmak@posta.mu.edu.tr
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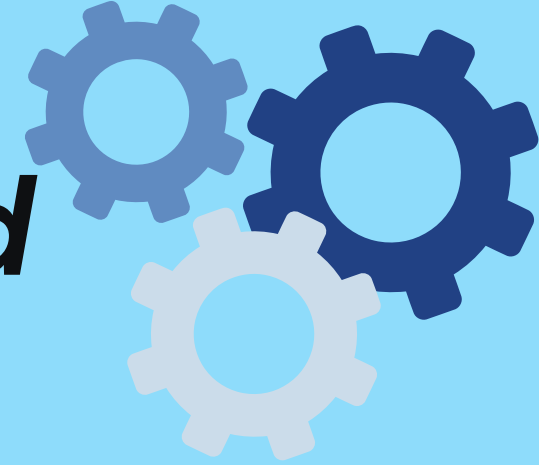
OUR MOTIVATION

With the rise of powerful generative models like GANs, AI-generated images have become increasingly realistic, posing challenges in verifying visual content across areas such as journalism, social media, and security. This project focuses on automatically detecting synthetic images to combat visual misinformation. We explore three detection strategies: classical machine learning with visual features, transfer learning using deep features, and state-of-the-art detection models. Our goal is to develop effective tools to help individuals and organizations assess image authenticity.



Initially, a different dataset was considered, but due to its lack of structure and documentation, we switched to the well-documented “140k Real and Fake Faces” dataset from Kaggle. It includes 70,000 real images from Flickr and 70,000 synthetic images generated by StyleGAN. All images are resized to 256×256 and pre-split into training, validation, and test sets with labeled CSV files. This diverse and high-quality dataset provides a solid foundation for training and evaluating face detection models.

First Approach: Traditional Feature-Based Detection



In this approach, nine handcrafted visual features (e.g., color histogram, edge density, HOG, LBP) were extracted from images. These features were used to train classical machine learning models like Random Forest, Logistic Regression, and MLP. The goal was to distinguish real and fake faces using interpretable, low-level image statistics.

- **Color Histogram**
- **Grayscale Statistics**
- **Edge Density**
- **Local Binary Pattern (LBP)**
- **Histogram of Oriented Gradients (HOG)**
- **Gabor Features**
- **Wavelet Features**
- **Image Moments**
- **Shannon Entropy**



Models	Total-Time	Accuracy	Precision	Recall	F1-Score
KNN	121,077	0.5668	0.5359	0.9916	0.6955
DECISION TREE	58,142	0.6760	0.6724	0.6853	0.6785
RANDOM FOREST	815,595	0.8257	0.8549	0.7846	0.8185
LOGISTIC REGRESSION	68,114	0.8831	0.8811	0.8856	0.8833
MULTI-LAYER PERCEPTION	210,804	0.9411	0.9394	0.9431	0.9412
XGBOOST	10,028	0.8971	0.8977	0.8963	0.8969
LIGHTGBM	80,051	0.8674	0.8767	0.8552	0.8645
CATBOOST	417,236	0.9251	0.9262	0.9237	0.9243



SECOND APPROACH: Deep Feature Extraction with Transfer Learning

This method uses a pre-trained ResNet50 model to extract deep features from images. Instead of training the entire network, only features from the last convolutional layer were used. These were fed into classical classifiers (e.g., MLP, Logistic Regression) to detect real vs. fake faces. This approach achieved the best overall performance with high accuracy and efficiency.



Models	Total-Time	Accuracy	Precision	Recall	F1-Score
KNN	35,489	0.7611	0.9279	0.5661	0.7032
DECISION TREE	91,591	0.7056	0.7087	0.6979	0.7033
RANDOM FOREST	199,082	0.8391	0.8419	0.8350	0.8384
LOGISTIC REGRESSION	12,281	0.9271	0.9297	0.9240	0.9268
MULTI-LAYER PERCEPTION	5,299	0.9544	0.9450	0.9648	0.9548
XGBOOST	23,492	0.9105	0.9143	0.9059	0.9101
LIGHTGBM	15,723	0.8882	0.8936	0.8814	0.8875
CATBOOST	255,352	0.9298	0.9348	0.9242	0.9295



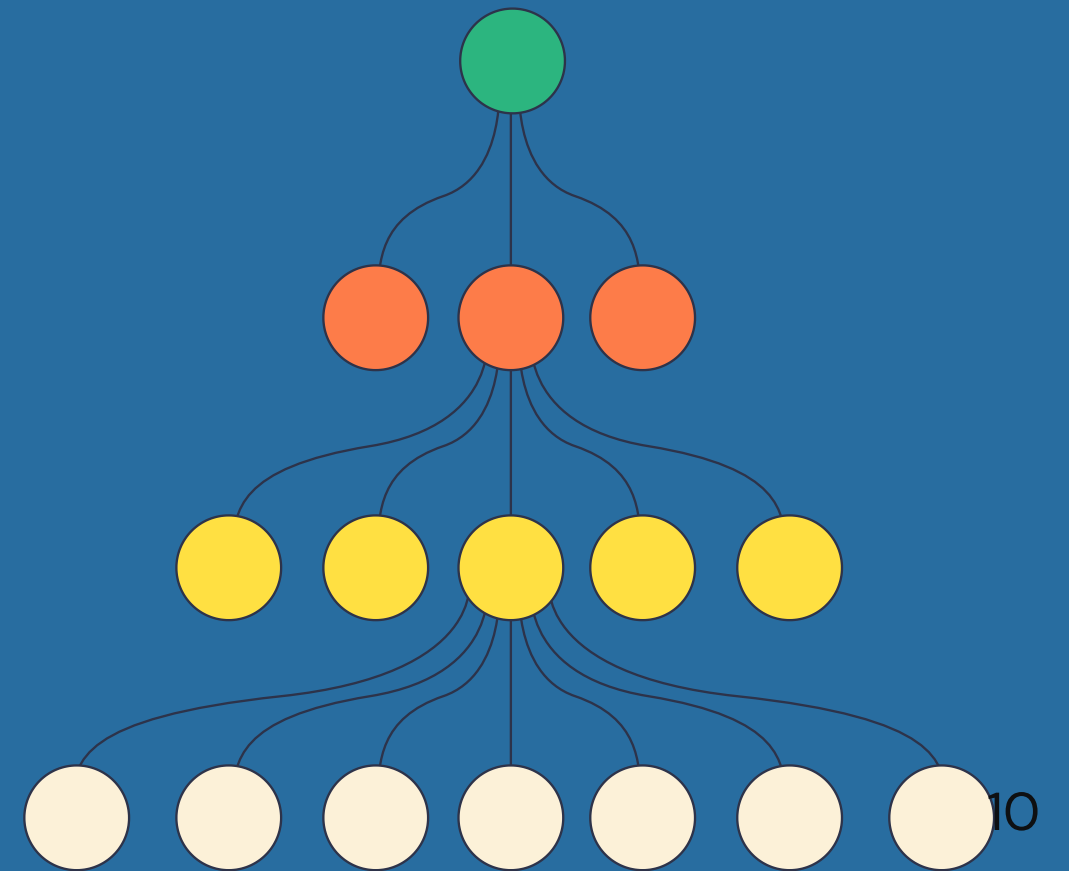
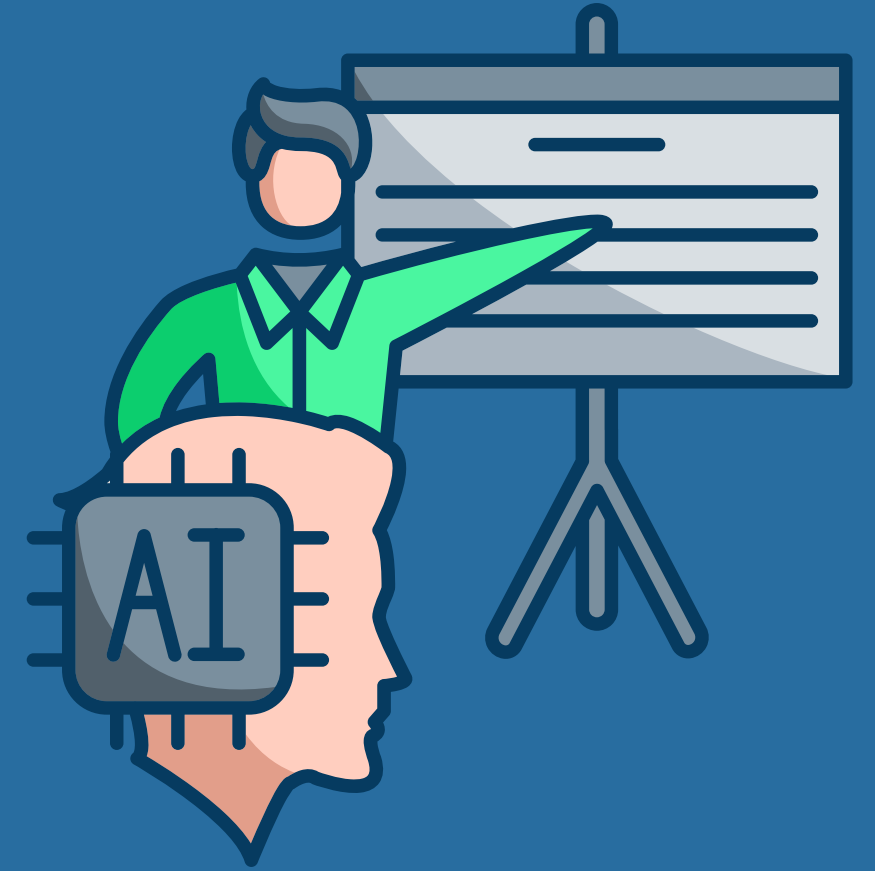
Third Approach: Benchmark Models for Synthetic Image Detection

This approach uses advanced, pre-trained models from the SIDBench benchmark (e.g., UnivFD, DIMD) designed specifically to detect GAN-generated images. These models can capture subtle artifacts in synthetic faces. While they showed very high precision, their recall was low and runtime was significantly longer, making them less practical for real-time use.

MODELS	Total-Time	Accuracy	Precision	Recall	F1-Score
UnivFD	11588 (3h)	0.8122	0.9692	0.3492	0.5134
DIMD	61435 (17h)	0.8923	0.9971	0.3149	0.4786

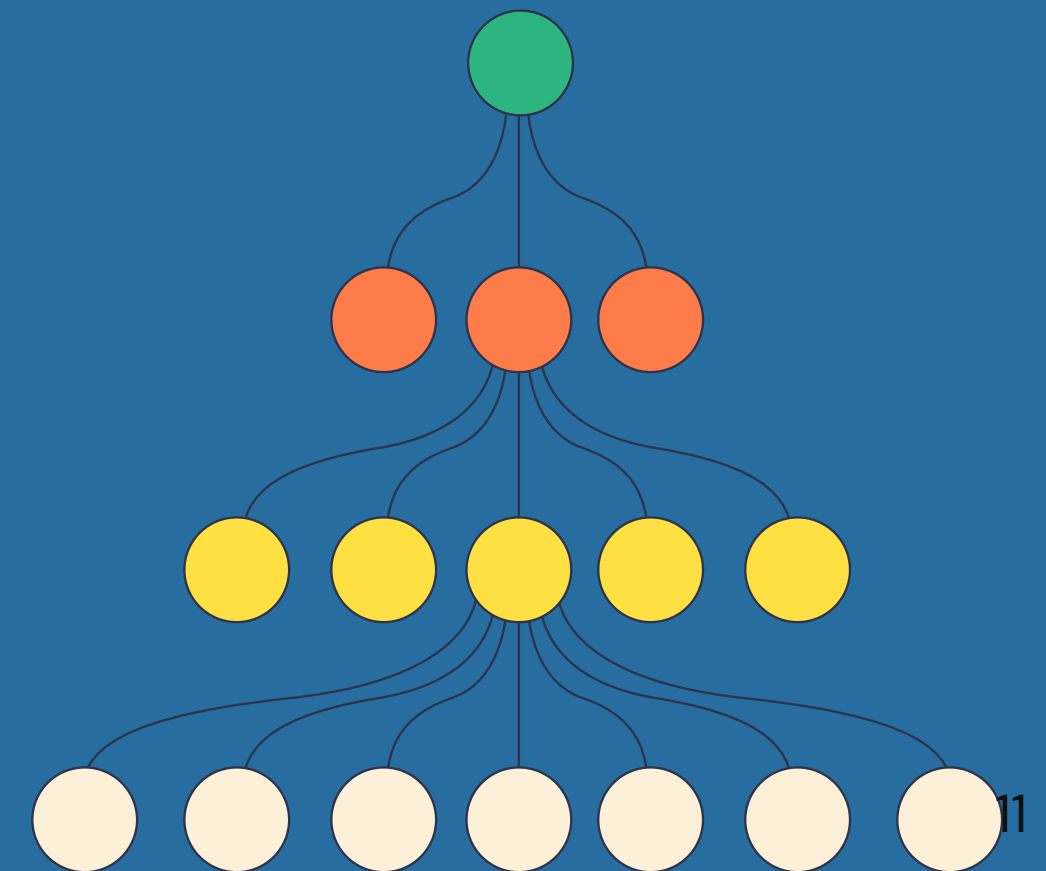
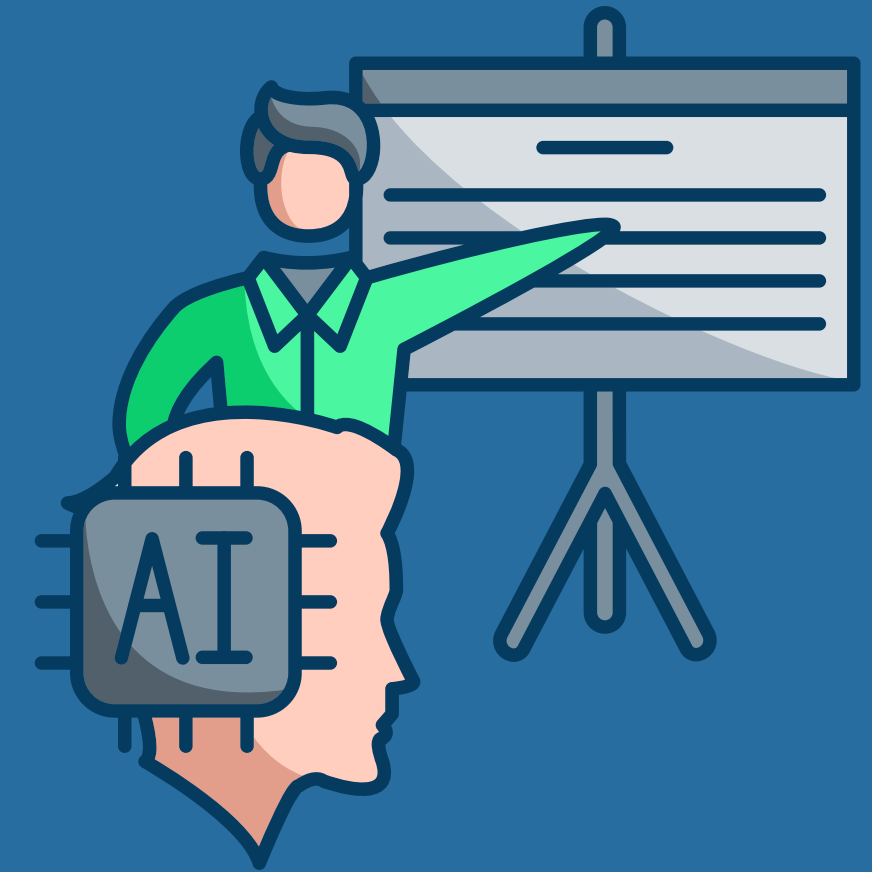
CONCLUSION

Among the three methods, the second approach (ResNet50 + classical classifiers) provided the best balance of accuracy and efficiency. The first approach, based on handcrafted features, was a solid alternative for limited-resource settings. The third approach, using benchmark models like UnivFD and DIMD, achieved high precision but suffered from low recall and long runtimes, making it less practical for real-time applications.



FUTURE STEPS

- *Use images from newer GANs, diffusion models, and deepfake tools.*
- *Test models on different datasets for better generalization.*
- *Apply hyperparameter tuning (e.g., grid search, Bayesian optimization).*
- *Build a user-friendly interface for non-technical users.*
- *Extend the system to detect AI-generated videos.*
- *Upgrade hardware (e.g., GPUs, TPUs) for faster training and testing.*



THANK YOU!

